Artificial Intelligence Lab

Lab Report # 03

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Dated:

March 14th, 2026

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Objectives: -

- To understand basics of Python Functions.
- Implementing Python Functions.
- Types of Passing Arguments and Return statement.

Question # 01

Write a Python function to calculate the factorial of a number (a non- negative integer). The function accepts the number as an argument.

Solution: -

```
script.py
def factorial(n):  1 usage
    if n<0:
        return "Invalid input: must be non-negative"
    result = 1
    for i in range(2, n + 1):
        result *= i
    return result
print(factorial(5))
```

Output: -

```
120

Process finished with exit code 0
```

Question # 02

Write a Python function that takes a list and returns a new list with unique elements of the

first list.

a) Sample List: [1,2,3,3,4,4,5,3,7,5]

b) Unique List: [1, 2, 3, 4, 5, 7].

Solution: -

```python
def unique_elements(lst):
    return list(set(lst))
sample_list = [1, 2, 3, 3, 4, 4, 5, 3, 7, 5]
print(unique_elements(sample_list))  # Output: [1, 2, 3, 4, 5, 7]
```

Output: -

```
[1, 2, 3, 4, 5, 7]

Process finished with exit code 0
```

Question # 03

Write a Python function to show the first ‘n’ even numbers and then return the sum and product of that even numbers outside the function.

Solution: -

```python
def get_even_numbers(n):  1 usage
    even_numbers = [i for i in range(2, 2 * n + 1, 2)]
    print("Even numbers:", even_numbers)
    return even_numbers
evens = get_even_numbers(5)
sum_evens = sum(evens)
product_evens = 1
for num in evens:
    product_evens *= num
print("Sum:", sum_evens)
print("Product:", product_evens)
```

Output: -

```
Even numbers: [2, 4, 6, 8, 10]
Sum: 30
Product: 3840

Process finished with exit code 0
```

Question # 04

Write a python program to match two string using functions

Solution: -

```
def match_strings(str1, str2):
    if str1 == str2:
        return "Strings match!"
    else:
        return "Strings do not match."
string1 = input("Enter first string: ")
string2 = input("Enter second string: ")
print(match_strings(string1, string2))
```

Output: -

```
Enter first string: AI is dangerous
Enter second string: AI is benefitial
Strings do not match.

Process finished with exit code 0
```

Question # 05

Write a python program to name user login system using function.

Solution: -

```python
import hashlib
def hash_password(password):  5 usages
    return hashlib.sha256(password.encode()).hexdigest()
def login_system(username, password, users_db):  1 usage
    hashed_input = hash_password(password)
    if username in users_db and users_db[username] == hashed_input:
        return "Login successful! Welcome."
    else:
        return "Login failed! Invalid username or password."
my_username = "ahmed"
my_password = "qwerty12@"
hashed_value = hash_password(my_password)
print(f"Hashed password for {my_username}: {hashed_value}")
users_db = {
    "admin": hash_password("12345"),
    "user1": hash_password("mypassword"),
    "alice": hash_password("securepass"),
    my_username: hashed_value
}
user = input("Enter username: ")
pwd = input("Enter password: ")
print(login_system(user, pwd, users_db))
```

Output: -

```
Hashed password for ahmed: c968aeb1c1d1cb94f08fe49b6492a1821df6c51ba3be7aee524f3b526b1ade78
Enter username: ahmed
Enter password: qwerty12@
Login successful! Welcome.

Process finished with exit code 0
```